

Project Master Overview: Robust and Explainable AI for Diabetic Retinopathy

Project Goal:

To develop a robust and explainable AI framework for Diabetic Retinopathy (DR) detection that maintains high diagnostic accuracy and interpretability across varying image quality conditions, leveraging Vision Transformers (ViTs), Explainable AI (XAI), and Generative AI (GenAI) for enhancement.

Core Research Questions (RQs):

1. **RQ1: Model Robustness to Degradation:** How do Vision Transformers compare to CNNs (ResNet-50, EfficientNet-B3) in DR classification performance when subjected to varying levels of synthetic image degradation (blur, exposure, noise) across balanced and unbalanced datasets?
2. **RQ2: XAI Fidelity & Stability under Degradation:** To what extent does the fidelity and stability of explainability methods (Grad-CAM, SHAP, and Self-Attention) degrade under different image quality conditions for CNNs and ViTs?
3. **RQ3: GenAI for Accuracy & XAI Recovery:** Can GenAI-based quality enhancement (e.g., Diffusion Models) significantly recover both diagnostic accuracy and XAI localization precision compared to traditional preprocessing (e.g., CLAHE) in low-quality fundus images?
4. **RQ4: Quality-Aware Ensemble for Clinical Trust:** How can a quality-aware ensemble framework dynamically select the optimal enhancement-model-XAI combination to maximize clinical trust in real-world DR screening?

Phased Methodology:

Phase 1: Data Engineering & Synthetic Degradation

- **Objective:** Create controlled degraded datasets to stress-test models.
- **Input Data:** "Good Quality" subset of APTOS 2019 and EyeQ datasets.
- **Process:** Apply three levels (Low, Medium, High) of synthetic degradation (Gaussian Blur, Exposure/Underexposure, Gaussian Noise) to pristine images.
- **Output:** Structured directories of degraded images, maintaining original labels.

Phase 2: Model Benchmarking (RQ1)

- **Objective:** Evaluate the robustness of CNNs and ViTs to image degradation.
- **Models:** Pre-trained ResNet-50, EfficientNet-B3, and a Vision Transformer (ViT-Base).
- **Process:** Train models on clean data. Evaluate performance (Accuracy, F1-Score, AUC-ROC) on the original clean test set and all synthetically degraded test sets.
- **Output:** Performance metrics and "Accuracy vs. Degradation Severity" plots for each model and degradation type.

Phase 3: Explainability Benchmark (RQ2)

- **Objective:** Quantify the degradation of XAI fidelity and stability under varying image quality.
- **XAI Methods:** Grad-CAM (for CNNs), SHAP (for all models), Attention Maps (for ViT).
- **Metrics:**
 - **Fidelity:** Deletion/Insertion Area Under the Curve (AUC).
 - **Localization:** Intersection over Union (IoU) with ground-truth lesion masks (if available).
 - **Stability:** Spearman Rank Correlation between XAI heatmaps of clean vs. degraded images.
- **Output:** "XAI Fidelity/Stability vs. Degradation" plots and tables.

Phase 4: GenAI Enhancement & XAI Recovery (RQ3)

- **Objective:** Assess GenAI's ability to restore both diagnostic accuracy and XAI interpretability.
- **Enhancement:** Apply a pre-trained GenAI model (e.g., Diffusion Model, CycleGAN) for fundus image denoising/deblurring to synthetically degraded images.
- **Evaluation:** Re-evaluate models (from Phase 2) on GenAI-enhanced images for accuracy. Re-calculate XAI metrics (from Phase 3) on enhanced images, comparing against original clean XAI.
- **Output:** "Accuracy Recovery" and "XAI Fidelity Recovery" plots/tables, demonstrating pathology preservation.

Phase 5: Quality-Aware Ensemble Framework (RQ4)

- **Objective:** Develop an adaptive system for real-world clinical deployment.

- **Process:** Integrate a quality classifier (e.g., EyeQ) to assess incoming images. Dynamically route images through optimal enhancement (GenAI/CLAHE) and classification models based on quality. Combine with robust XAI for final diagnosis and explanation.
- **Output:** A conceptual framework and preliminary implementation of a quality-aware, interpretable DR screening pipeline.

Technical Requirements for Agent Implementation:

- **Libraries:** PyTorch, torchvision, Pillow, NumPy, Captum (or similar XAI library).
- **Data Access:** Assume data is accessible via mounted Google Drive paths (e.g., `/content/drive/MyDrive/`).
- **Output:** All results (metrics, plots, enhanced images, XAI heatmaps) should be saved in structured directories and formats (CSV, PNG, JSON) for easy analysis and reporting.
- **Code Structure:** Modular Python scripts for each phase, with clear function definitions and comments.

This overview provides a complete context for the project, enabling an AI agent to understand the scientific objectives and implement the technical steps effectively.